Mend-Amar Badral

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Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop solutions for the real-world, impactful problems.

Education

Budapest University of Technology and Economics (BME) — Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering — Budapest, Hungary
Supervisor: Márton Vaitkus
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) — Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics — Ulaanbaatar, Mongolia
GPA: 3.2/4.0 **(rank 3/34)**
Supervisor: Galtbayar Artbazar
Thesis: *Evaluating land degradation using image processing methods*

Experience

Center of Mathematics Application, NUM — Oct, 2023 – June, 2025
Undergraduate Research Intern

- **Lab Lead:** Selected as Undergraduate Lab Lead amongst the 6-8 student team in the final year. Co-taught and discussed a machine learning topics in weekly seminar meetings.
- **Tool development:** Developed a custom, vendor specific **software** which takes in drone camera images and sensor parameters to calculate soil adjusted vegetation index. Made a command line interface for the tool such that it can process images in custom folder structure in Linux environment.
- **Organization:** Implemented a unified, documented folder structure for the research data backup which enabled efficient data retrieval for research purposes and easy integration of data to be collected in the future.
- **Contributions:** Incorporated an open-source Docker containerized application (OpenDroneMap) for drone orthomosaic processing instead of proprietary software. Enabled fast research iteration cycles by analyzing and performing comparisons on the results for discussion with the supervisor for feedback.

Projects

aiSim Data Integration for 3D Gaussian Splatting

- Implemented a JSON parser to convert aiSim, a simulation software, into a 3D scene representation software called nerfstudio's format.

Deep Learning Foundations and Concepts Book Examples Reproduction

- Reproduced examples and figures from C. Bishop's *Deep Learning Foundations and Concepts* book.

Skills

Programming: Python, Julia, C/C++, R, Bash, MATLAB
Tools: Docker, Linux, Git, LaTeX